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RAJALAKSHMI
INSTITUTE OF
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GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 3 of 6

Innovation and Creativity

What creativity is and how it is measured · barriers and creative confidence · invention and innovation · four types of innovation · where ideas come from

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In this presentation

- What creativity actually means
- Four measures of creative output
- Barriers to creativity
- Creative confidence
- Invention and innovation
- Four types of innovation
- Where ideas actually come from
- In class: the thirty circles

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Creativity

What it is, how it is measured, and what gets in its way

What creativity actually means

- **Novel and useful** together – novelty on its own is not creativity
- A skill that improves with method and practice, not a fixed trait you are born with
- Most often recombination: connecting two things that were already there
- Everyone is creative in a domain they know well; the method is what transfers

KEY POINT New but useless is novelty. Useful but familiar is routine. Creativity needs both at once.

Four measures of creative output

Fluency

How many ideas you produced. The first measure, and the one most people stop short on.

Flexibility

How many different kinds of idea. Twenty variations of one idea score low here.

Originality

How unusual the idea is compared with what everyone else in the room produced.

Elaboration

How far an idea has been developed – detail, consequences, how it would work.

Barriers to creativity

- **Perceptual** – seeing the problem in only one way
- **Emotional** – fear of being wrong or of being judged
- **Cultural** – “we have always done it like this”
- **Environmental** – no time, no space, the boss in the room
- **Intellectual** – reaching for the familiar method first
- **Expressive** – having the idea but being unable to show it

Creative confidence

- The belief that you can change the things around you – and the willingness to try
- Most people lose it at school, when drawing and guessing begin to be marked
- It returns through small wins: a rough sketch someone understood, a prototype someone used
- Guided failure in a safe setting rebuilds it – which is what this course is for

KEY POINT You do not need to draw well. You need to be willing to be seen drawing badly.

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Innovation

Turning an idea into something that reaches people

Invention and innovation

Invention

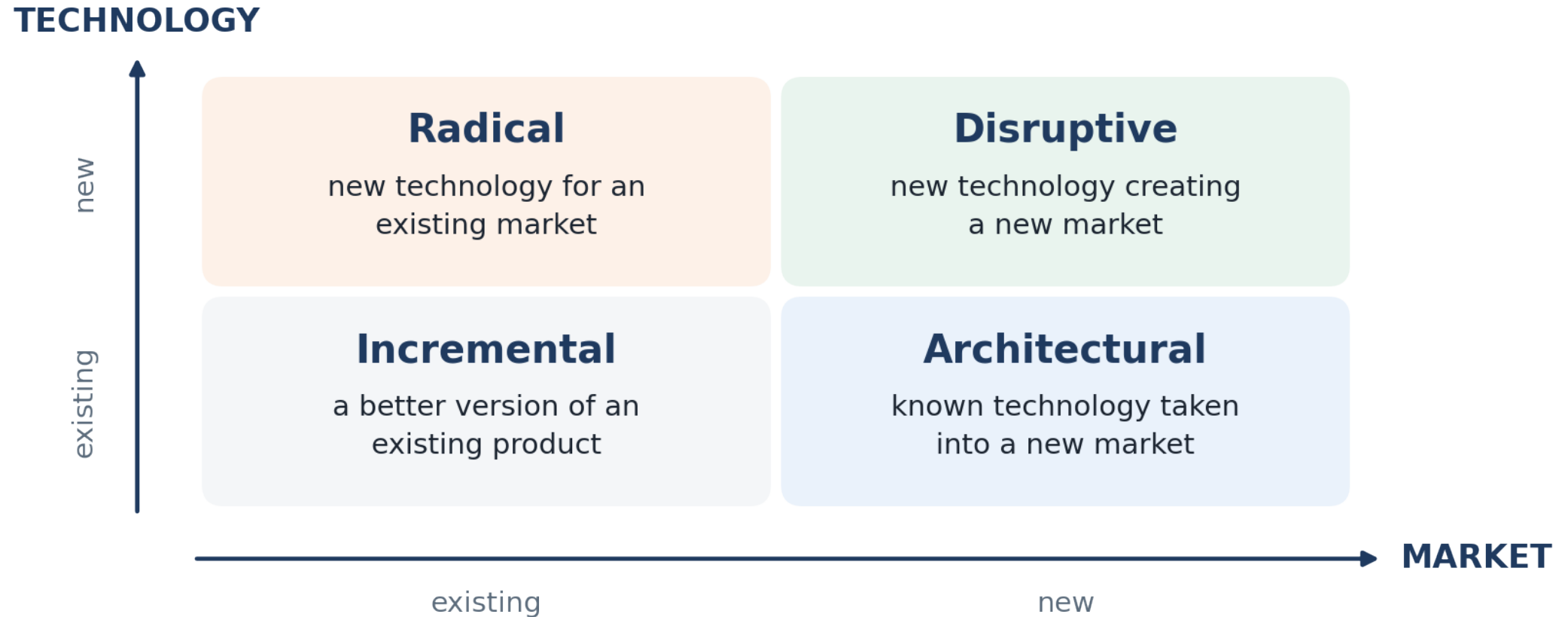
- A new idea, device or method
- Proven in the laboratory
- Measured by novelty and patents
- Can sit unused for decades

Innovation

- An invention put into real use
- Proven in the market or field
- Measured by adoption and impact
- Needs a model, a chain, a user

KEY POINT Invention costs money. Innovation is what makes the invention matter.

Four types of innovation



Most engineering work is incremental – design thinking is just as useful there.

The four types, with examples

Type	What is new	Example
Incremental	Neither	Each new model of a mixer grinder
Architectural	The market	A clay pot sold as a household refrigerator
Radical	The technology	LED lamps replacing filament bulbs
Disruptive	Both	A portable infant warmer for home births

Design thinking is not reserved for the bottom row. It is how the top row stays worth buying.

Where ideas actually come from

- **Analogy** – how is this solved in a completely different field?
- **Constraint** – what would you do if it had to cost one tenth as much?
- **Extremes** – how would the most demanding user solve it?
- **Combination** – what happens if two existing things are joined?
- **Reversal** – what if the usual order were inverted?

KEY POINT None of these needs inspiration. They are prompts you can apply on purpose, in any order.

In class: the thirty circles

- You get a sheet of thirty empty circles and three minutes
- Turn as many as you can into recognisable objects
- Count your fluency (how many) and your flexibility (how many kinds)
- Then compare: did anyone join circles, or draw outside the sheet?

ACTIVITY Three minutes, alone and in silence. Do not stop to judge an idea while the clock is running.

THINK ABOUT IT

**Nobody was told the circles had to stay separate.
Why did almost everyone keep them that way?**

- Which rule did you assume existed but was never stated?
- Where else do you follow a rule nobody actually set?
- What would you have drawn if only the count mattered?
- How would you catch this assumption in a real project?

Summary

- Creativity is novelty plus usefulness – and it is a trainable skill
- It is measured by fluency, flexibility, originality and elaboration
- Six barriers block it; creative confidence is rebuilt by small wins
- Invention is a new idea; innovation is that idea put to use
- Four types: incremental, architectural, radical, disruptive
- Ideas can be prompted on purpose by analogy, constraint and reversal

Next: The design thinking process

- The five stages, one at a time
- IDEO's three spaces of innovation
- The Double Diamond and the five-stage model compared
- What every model of the process agrees on